

SECTION 15831-A
AIR-DISTRIBUTION AND - EXHAUST FANS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes utility sets, centrifugal fans, propeller fans, roof fans, and vaneaxial fans for air distribution systems.

1.3 REFERENCES

- A. Conform to AMCA bulletins regarding construction and testing. Fans shall bear AMCA certified rating seal.

1.4 SUBMITTALS

- A. Submit Shop Drawings and Product Data in accordance with Division 1.
- B. Submittals of fan selection and performance shall show operating point plotted on fan curves. Where backdraft dampers are included, the plotted performance shall incorporate the resistance of the backdraft dampers.
- C. Submit backdraft damper and motorized shutoff damper pressure drop data.
- D. Submit fan unit inlet, outlet and casing-radiated octave band sound power levels (dB re: 10^{-12} watt) and dynamic balance data. See Section 15240-Mechanical Vibration Control.
- E. Submit manufacturer's printed installation instructions.
- F. Submit operating and maintenance manuals.
- G. Submit installation and start-up certification.
- H. Submit vibration isolation details, cut sheets, calculations, and drawings in accordance with Section 15240.

1.5 QUALITY ASSURANCE

- A. Units shall be products of manufacturer regularly engaged in production of such units and issuing complete catalog data on such products.
- B. Manufacturer shall assure that all equipment provided does not exceed maximum allowable vibration peak-to-peak displacement data listed in the latest ASHRAE Systems and Applications Handbook and the requirements specified in Section 15240.

1.6 JOB CONDITIONS

- A. Do not operate fans for any purpose, temporary or permanent, until ductwork is clean, filters are in place, bearings have been lubricated, and fan has been run under observation.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Equivalent fan selections shall not increase motor horsepower, increase noise level, increase tip speed by more than 10 percent, or increase inlet air velocity by more than 20 percent, from that scheduled.
- B. Base fan performance on altitude rating conditions.
- C. Refer to Drawings for performance requirements.

2.2 MOTORS AND CONTROLS

- A. Motors: Provided with equipment. Refer to Section 15010. Motors for the General Exhaust fans shall be TEFC.
- B. Motor Starters: Refer to Section 15010.

2.3 SELECTION AND BALANCING

- A. Static Pressure Variations: Provide fans capable of accommodating static pressure variations of plus 10 percent. Fans shall be capable of overcoming losses of backdraft dampers, where specified.
- B. Fan Motor Brake Horsepower: Shall include air performance drive losses, inlet vane losses and all other losses of related manufacturer furnished components as

applicable.

- C. Sheaves: Provide balanced variable sheaves for motors 15 HP and under, and fixed sheaves for 20 HP and over.
- D. Balancing: Statically and dynamically balance fans to eliminate vibration or noise transmission to occupied areas of the building. See Section 15240 for balance requirements
- E. Belt and Shaft Guards: Galvanized steel sides and expanded metal face with opening for tachometer for belts and sheaves, galvanized steel for exposed drive shafts.
- F. Safety Screen: Provide where inlet or outlet is exposed.
- G. Weather Hood: Provide for fan motors and drives exposed to the weather.

2.4 PAINTING

- A. Factory Prime Coat: Required on all surfaces of ferrous fan housings and wheels. Painting outside is not required on aluminum fans.
- B. Ferrous Fans Exposed to Weather: Prime coat and finish with corrosion resistant epoxy paint, color selected by Architect.
- C. General Exhaust (GEX) System Fans: Factory prime coat and finish with corrosion resistant epoxy paint of standard manufacturer color.

2.5 CENTRIFUGAL FANS

- A. General: Provide multiblade wheels, reinforced steel housing and mounting base, access door and drain connection in scroll. Housings shall be continuously welded steel construction rated and tested in accordance with AMCA Standards. Wheel shall be backward inclined airflow design with non overloading characteristics. The wheel shall be statically and dynamically balanced to achieve the vibration criteria as required in Section 15240. Provide flanges on the fan inlet and outlet.
- B. Drive: Direct drive or multiple V-belt, as specified in Section 15240, adjustable motor mount, heavy duty, self-aligning, spherical roller type bearings with external lubrication fittings, 200,000 hour L-50 life, sheaves, and steel shaft.

- C. Variable Volume Inlet Vanes: Provide where scheduled.
- D. Vibration Isolation: Refer to Section 15240.

2.6 POWERED ROOF VENTILATORS

- A. General: Provide centrifugal fans, reinforced aluminum housings, up-blast type with drain trough or down-blast. Refer to Drawings.
- B. Drives: V-belt, adjustable motor mount supported from main fan housing through neoprene vibration isolators and provided with extended lubrication fittings. Use direct driven fans only where specifically scheduled as such.
- C. Accessories: Provide birdscreen, backdraft damper, (multiblade, felt-edged, rattle free, backdraft damper), integral curb cap, motorized shut off damper.
- D. Provide vibration isolation as specified in Section 15240.

2.7 UTILITY SETS

- A. General: Provide multiblade wheels, reinforced steel housing and mounting base, access door and drain connection in scroll. Provide blade type and discharge arrangements as scheduled and indicated on Drawings.
- B. Drive: V-belt, adjustable motor mount, heavy-duty self-aligning bearings, extended lubrication fittings, sheaves, and steel shaft.
- C. Vibration Isolation: Refer to Section 15240.

2.8 VANEAXIAL FANS

- A. General: Provide adjustable diecast impeller blades with motor as an integral part of the hub design, flanged housing for connection to ductwork, and easy access to fan wheel for varying blade angle setting during air balancing. Extend lubrication fittings outside of housing.
- B. Accessories: Provide junction box. Provide inlet vanes where fans are not duct connected.
- C. Variable Inlet Vanes: Provide where scheduled.
- D. Vibration isolation see Section 15240.

2.9 PLUG FANS

- A. Fans: Steel, single width, single inlet, plug type with multiblade wheel; keyed to a turned, ground and polished steel shaft; statically and dynamically balanced according to Section 15240, V-belt driven, zinc chromate primed, and finished with baked enamel. Refer to schedule on Drawings for specific fan requirements and arrangement.
- B. Bearings: Self-aligning pillow block regreaseable ball type, 200,000 hour average life per ANSI B3.15. Extend lubrication fittings to accessible locations exterior of fan casings.
- C. Variable Sheaves: Provide for motors 15 HP and smaller and fixed sheaves for 20 HP and larger.
- D. Motor Mount: Adjustable slide type.
- E. Belt and Shaft Guards: Galvanized steel sides and an expanded metal face with opening for tachometer for belts and sheaves, galvanized steel for exposed drive shafts.
- F. Variable Inlet Vanes: Provide conical inlet vanes with zero pressure loss in full open position, where scheduled.
- G. Vibration isolation see Section 15240.

2.10 ROOF MOUNTED INTAKE AND RELIEF HOODS

- A. General: Provide reinforced aluminum housing with integral throat, birdscreen, and curb cap. Refer to Drawings. Greenheck Model WRH, or equal.
- B. Accessories: Provide matching aluminum curb with integral cants, nail strips, and rigid fiberglass insulation.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Supply and install sheaves as necessary for final air balancing. Install fans in accordance with manufacturer's instructions.
- B. Set roof-mounted fans on curbs. Provide drip pan for

collecting condensation as required.

C. Pipe drainage from centrifugal fans to drain.

3.2 START-UP

A. Inspect equipment after installation to verify installation is in accordance with specifications, equipment is lubricated, proper belt tension, and that equipment is otherwise ready to operate.

B. Perform air side test and balance as applicable. Refer to Section 15950.

C. Provide the services of manufacturer's representative for starting up units and instruction of Owner's operating personnel.

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